THE NAIVE COTANGENT COMPLEX OF AN AFFINE CHART COMPUTES THAT OF THE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3735-3745 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3747-3774, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The naive cotangent complex of a ring map and of a morphism, presentations, and sheafification; the comparison is retained.

Source reference. Original lemma 07BS: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3774–3774. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00S7: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3774–3774. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D09: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3774–3774. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08TG: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3774–3774. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01I7: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3774–3774. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01HV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3774–3774. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 008H: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3774–3774. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $\mathrm{Spec}(A) = U \subset X$ and $\mathrm{Spec}(R) = V \subset S$ be affine opens with $f(U) \subset V$. There is a canonical map*

$$\widetilde{NL_{A/R}} \longrightarrow NL_{X/Y}|_U$$

of complexes which is an isomorphism in $D(\mathcal{O}_U)$.

Proof. From the construction of $NL_{X/Y}$ in Modules, Section 08TG we see there is a canonical map of complexes $NL_{\mathcal{O}_X(U)/f^{-1}\mathcal{O}_Y(U)} \to NL_{X/Y}(U)$ of $A = \mathcal{O}_X(U)$-modules, which is compatible with further restrictions. Using the canonical map $R \to f^{-1}\mathcal{O}_Y(U)$ we obtain a canonical map $NL_{A/R} \to NL_{\mathcal{O}_X(U)/f^{-1}\mathcal{O}_Y(U)}$ of complexes of A-modules. Using the universal property of the $\widetilde{\ }$ functor (see Schemes, Lemma 01I7) we obtain a map as in the statement of the lemma. We may check this map is an isomorphism on cohomology sheaves by checking it induces isomorphisms on stalks. This follows from Algebra, Lemma 07BS and 00S7 and Modules, Lemma 0D09 (and the description of the stalks of $\mathcal{O}_X$ and $f^{-1}\mathcal{O}_Y$ at a point $\mathfrak{p} \in \mathrm{Spec}(A)$ as $A_{\mathfrak{p}}$ and $R_{\mathfrak{q}}$ where $\mathfrak{q} = R \cap \mathfrak{p}$; references used are Schemes, Lemma 01HV and Sheaves, Lemma 008H). □